SME206
Signals and Systems

3. Fourier Series Representation of Periodic Signals

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Previously in Chapter 2…

- **Signals** represented by the sums of scaled and time-shifted impulse functions

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k] \qquad x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$$

- For **LTI systems**, the **output** is thus represented by the sum of scaled and time-shifted impulse responses (using the superposition property)

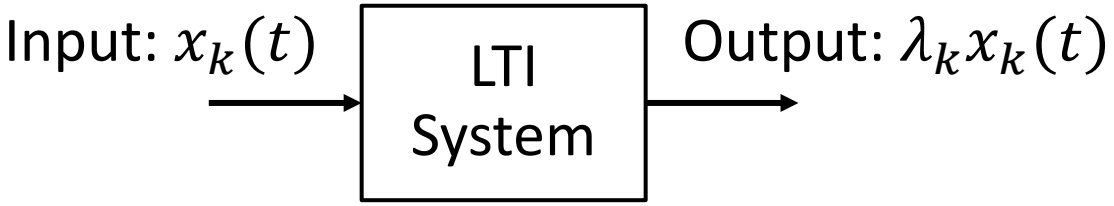
$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] \qquad x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

- In this chapter, **signals** decomposed into sums of scaled complex exponential functions, with some nice behavior in LTI systems.

Eigenfunctions of an LTI System

- For an **LTI system**, if the output is a *scaled* version of its input, then the input function is called an **eigenfunction** of the system.
- The scaling factor is called the **eigenvalue** of the system.

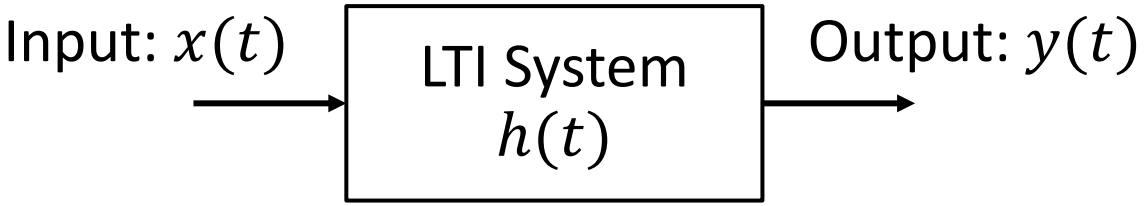
In other words, identify a family of signals $\{x_k(t)\}$ such that:



where λ_k is a scale factor

The Response of LTI Systems to Complex Exponentials (CT)

- Consider an LTI system with impulse response $h(t)$ and input signal $x(t)$:



- Suppose that $x(t) = e^{st}$ for some $s \in C$, we have for output $y(t)$:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau = \int_{-\infty}^{+\infty} h(\tau)e^{s(t-\tau)}d\tau = e^{st}\int_{-\infty}^{+\infty} h(\tau)e^{-s\tau}d\tau$$

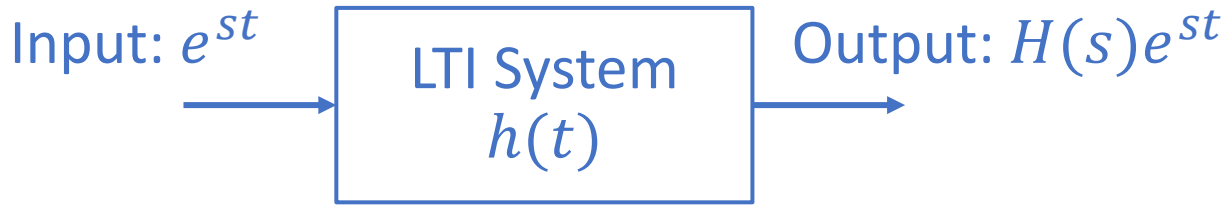
- Define $H(s) = \int_{-\infty}^{+\infty} h(\tau)e^{-s\tau}d\tau$, assuming this integral is finite (i.e., $H(s)$ is just some complex number). We have:

$$\boldsymbol{y(t) = H(s)e^{st}}$$

  - $\boldsymbol{H(s)}$**: Transfer Function** of the CT LTI system, defined by the impulse response $h(t)$.
  - $H(s)$ is independent of t, and is thus a scalar multiplied to the function $x(t)$.

The Response of LTI Systems to Complex Exponentials (CT)

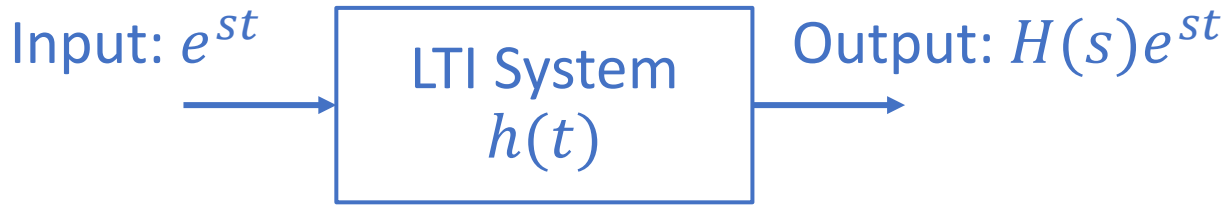
- We see that for an LTI system, if input $x(t) = e^{st}$ is applied, we have the output $y(t) = H(s)e^{st}$.



- $\boldsymbol{e^{st}}$ is called an **eigenfunction** of the continuous-time LTI system
- $\boldsymbol{H(s)}$ is the corresponding **eigenvalue**.

The Response of LTI Systems to Complex Exponentials (CT)

- We see that for an LTI system, if input $x(t) = e^{st}$ is applied, we have the output $y(t) = H(s)e^{st}$.



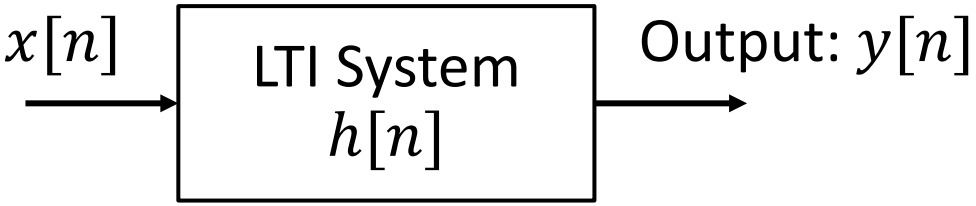
- **Special case**
  - Consider the periodic complex exponentials $e^{j\omega t}$, $\omega \in R$ (*i.e.*, $s = j\omega$), we have:

$$H(s) = H(j\omega) = \int_{-\infty}^{+\infty} h(\tau)e^{-j\omega\tau}d\tau$$

  - $H(j\omega)$ is called the **frequency response** of the system.

The Response of LTI Systems to Complex Exponentials (DT)

- Consider a DT LTI system with impulse response $h[n]$ and input signal $x[n]$:



- Suppose that $x[n] = z^n$ for some $z \in C$, we have for output $y[n]$:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} h[k]x[n-k] = \sum_{k=-\infty}^{+\infty} h[k]z^{n-k} = z^n \sum_{k=-\infty}^{+\infty} h[k]z^{-k} = H(z)z^n$$

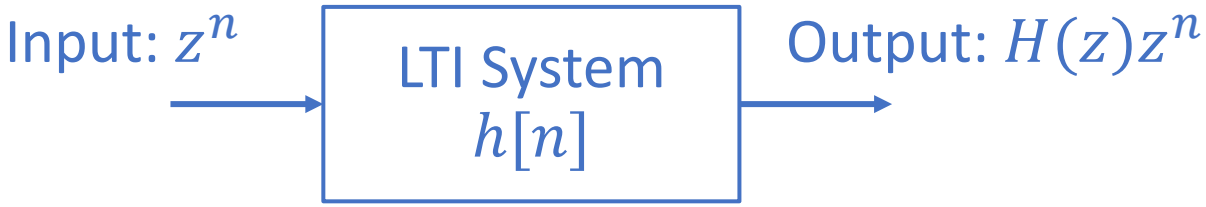
- Define $H(z) = \sum_{k=-\infty}^{+\infty} h[k]z^{-k}$. We have:

$$\boldsymbol{y[n] = H(z)z^n}$$

  - $\boldsymbol{H(z)}$: **Transfer Function** of the DT LTI system.

The Response of LTI Systems to Complex Exponentials (DT)

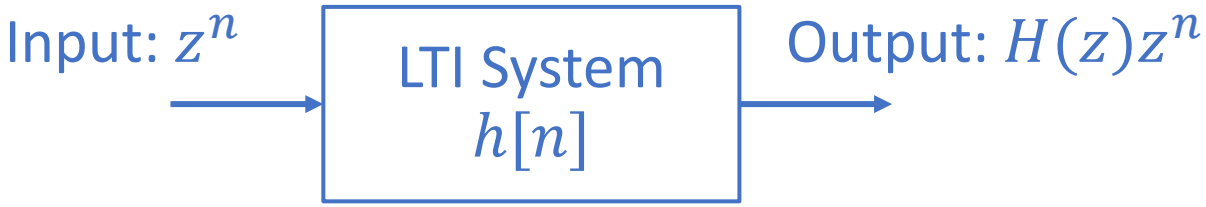
- We see that for an LTI system, if input $x[n] = z^n$ is applied, we have the output $y[n] = H(z)z^n$.



- $\boldsymbol{z^n}$ is called an **eigenfunction** of the discrete-time LTI system
- $\boldsymbol{H(z)}$ is the corresponding **eigenvalue**.

The Response of LTI Systems to Complex Exponentials (DT)

- We see that for an LTI system, if input $x[n] = z^n$ is applied, we have the output $y[n] = H(z)z^n$.



- **Special case**
  - Consider the periodic complex exponentials $e^{j\left(\frac{2\pi}{N}\right)n}$ (*i.e.*, $z = e^{j2\pi/N}$), we have:

$$H(z) = H\left(e^{j\Omega}\right) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\Omega k} \quad \text{, where } \Omega = 2\pi/N$$

  - $H(e^{j\Omega})$ is called the **frequency response** of the system.

Linear Combination of Signals

- Let $x(t)$ correspond to a linear combination of three complex exponentials:

$$x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + a_3 e^{s_3 t}$$

- From the *eigenfunction property*, the LTI response to each complex exponential is:

$$e^{s_1 t} \longrightarrow H(s_1) e^{s_1 t}$$
$$e^{s_2 t} \longrightarrow H(s_2) e^{s_2 t}$$
$$e^{s_3 t} \longrightarrow H(s_3) e^{s_3 t}$$

- From the *linear property* of an LTI system, we have the output $y(t)$:

$$\boldsymbol{y(t) = a_1 H(s_1) e^{s_1 t} + a_2 H(s_2) e^{s_2 t} + a_3 H(s_3) e^{s_3 t}}$$

→ If the input is a linear combination of complex exponentials,
the output of an **LTI system** is also a linear combination of complex exponentials.

Linear Combination of Signals

- **More generally:**
  - if the input to an LTI system is a linear combination of complex exponentials, then the output is also a linear combination of the same complex exponentials.

$$x(t) = \sum_k a_k e^{s_k t} \longrightarrow y(t) = \sum_k a_k H(s_k) e^{s_k t}$$

CT LTI

$$x[n] = \sum_k a_k z_k{}^n \longrightarrow y[n] = \sum_k a_k H(z_k) z_k{}^n$$

DT LTI

Linear Combination of Signals

Q. Consider an LTI system: $y(t) = x(t-3)$, where the input to this system is the complex exponential signal $x(t) = e^{j2t}$.

1. By direct substitution, what is the output $y(t)$?
2. By Eigenfunction property,
   1) What is the transfer function $H(s)$?
   2) What is the output $y(t)$?

3. Are the $y(t)$ obtained by the two above two approaches equivalent?

Linear Combination of Signals

Q. Consider the signal $x(t) = \cos(\omega t)$. Suppose that the impulse response of the system is real-valued.

1. What is the input $x(t)$ in the form of complex exponentials?
2. Consider that $h(t) \in R$ for all t, find the relation between $H(-j\omega)^*$ and $H(j\omega)$.
3. By Eigenfunction property, what is the output $y(t)$?

Self-Practice: Now, consider the system $y(t) = x(t - t_0)$, where $t_0 \in R$. Suppose we apply the signal $x(t) = \cos(\omega_0 t)$.

1. Calculate the output $y(t)$ by direct substitution.
2. Calculate the output $y(t)$ by the Eigenfunction property.
3. Compare the results obtained in 1 and 2.

Fourier Series Representation of CT Periodic Signals

- Consider the *periodic complex exponential signal*: $x(t) = e^{j\omega_0 t}$
  - Periodic with fundamental frequency ω_0 and fundamental period $T = 2\pi/\omega_0$.
- *Recall: harmonically related complex exponentials*

$$\boldsymbol{\phi_k(t) = e^{jk\omega_0 t} = e^{jk(2\pi/T)t}, \qquad k = 0, \pm 1, \pm 2, \ldots}$$

  - Each of these signals has a fundamental frequency: multiple of ω_0
  - Each is periodic with a period T (Not necessarily fundamental period!):
    - e.g., for $|k| \geq 2$, the fundamental period of $\phi_k(t)$ is a fraction of T.
  - Thus, a linear combination of these signals are also periodic with a period T.

Fourier Series Representation of CT Periodic Signals

- Consider a **linear combination of the harmonically related complex exponentials**:

$$x(t) = \sum_{-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$$

  - This representation is called the **Fourier Series representation of $x(t)$**.
  - The quantities, $a_k, k \in Z$ are called the **Fourier Series coefficients**.
  - The term for $k = 0$: a constant → known as *DC component (or DC term)*
  - The terms for $k = +1$ and $k = -1$: fundamental frequency of ω_0
    → known as the *fundamental components* or the *first harmonics*
  - The terms for $k = +2$ and $k = -2$: periodic with half the period (*i.e.*, $2\omega_0$) of the fundamental components → known as the *second harmonics*
  - Generally, the components for $k = +N$ and $k = -N$ are known as the *Nth harmonic components (Nth harmonics)*

Fourier Series Representation of CT Periodic Signals

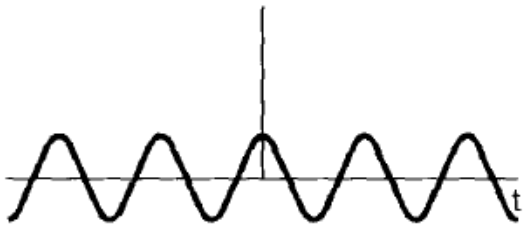
- Example: $x(t) = \sum_{k=-3}^{+3} a_k e^{jk2\pi t}$, where

$$a_0 = 1, \qquad a_1 = a_{-1} = 1/4\,, \qquad a_2 = a_{-2} = 1/2\,, \qquad a_3 = a_{-3} = 1/3$$

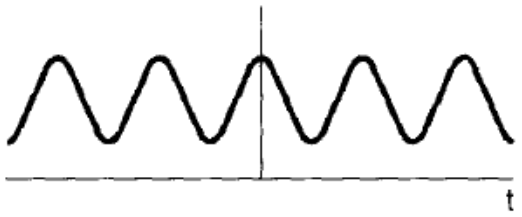
- $x(t) = 1 + \frac{1}{4}\left(e^{j2\pi t} + e^{-j2\pi t}\right) + \frac{1}{2}\left(e^{j4\pi t} + e^{-j4\pi t}\right) + \frac{1}{3}\left(e^{j6\pi t} + e^{-j6\pi t}\right)$
- By Euler's relation, we have
- $x(t) = 1 + \frac{1}{2}cos2\pi t + \cos 4\pi t + \frac{2}{3}cos6\pi t$

Fourier Series Representation of CT Periodic Signals

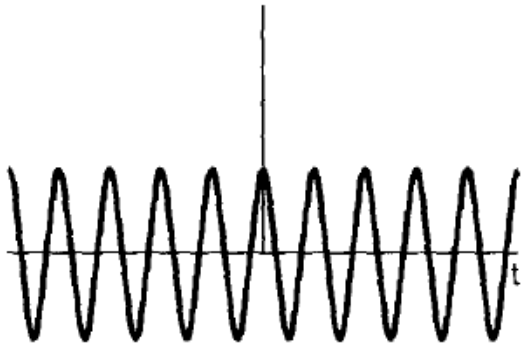
$x_1(t) = \frac{1}{2}\cos 2\pi t$



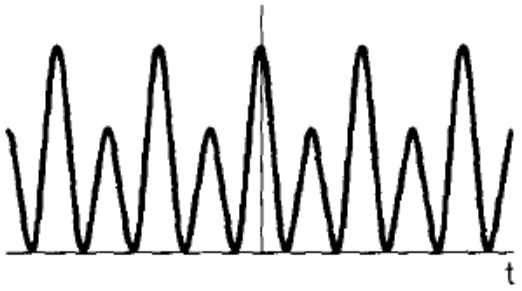
$x_0(t) + x_1(t)$



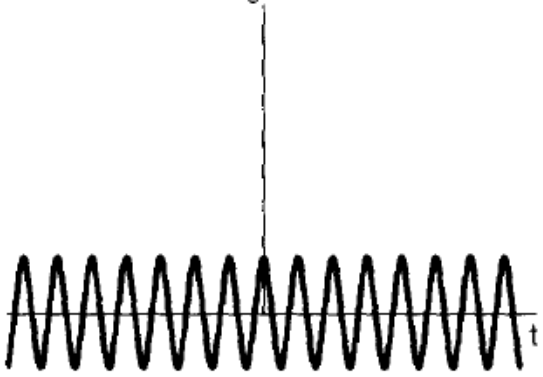
$x_2(t) = \cos 4\pi t$



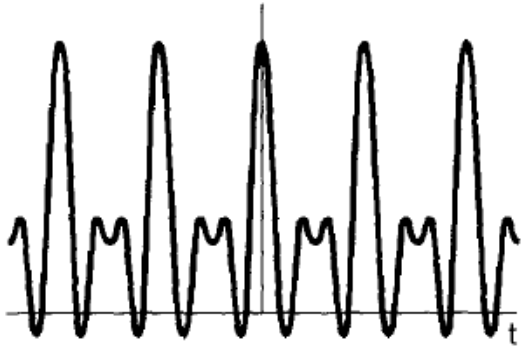
$x_0(t) + x_1(t) + x_2(t)$



$x_3(t) = \frac{2}{3}\cos 6\pi t$



$x(t) = x_0(t) + x_1(t) + x_2(t) + x_3(t)$



Determination of the FS Representation (CT)

- Suppose that $x(t)$ is periodic with period T, and that it has a FS representation:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

- How to find the coefficients $a_k, k \in Z$?
- Multiplying $x(t)$ by $e^{-jn\omega_0 t}$, we have:

$$x(t)e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t}$$

- Integrating both sides from 0 to $T = 2\pi/\omega_0$ (here, T is the fundamental period):

$$\int_0^T x(t)e^{-jn\omega_0 t}dt = \int_0^T \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t}\, dt$$

Determination of the FS Representation (CT)

$$\int_0^T x(t)e^{-jn\omega_0 t}dt = \int_0^T \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}e^{-jn\omega_0 t}\,dt = \sum_{k=-\infty}^{\infty} a_k \left[\int_0^T e^{j(k-n)\omega_0 t}dt\right]$$

- By Euler's relation, we have:

$$\int_0^T e^{j(k-n)\omega_0 t}dt = \int_0^T \cos\big((k-n)\omega_0 t\big)dt + j\int_0^T \sin\big((k-n)\omega_0 t\big)dt = \begin{cases} T, & k = n \\ 0, & k \neq n \end{cases}$$

- Thus,

$$\int_0^T x(t)e^{-jn\omega_0 t}dt = Ta_n \quad \Longrightarrow \boldsymbol{a_n = \frac{1}{T}\int_0^T x(t)e^{-jn\omega_0 t}dt}$$

- In fact, we obtain the same result if we integrate over *any interval of length T*:

$$\boldsymbol{a_n = \frac{1}{T}\int_T x(t)e^{-jn\omega_0 t}dt}$$

Determination of the FS Representation (CT)

- **Summary**: **If $x(t)$ has a Fourier series representation** (*i.e.,* if it can be expressed as a linear combination of harmonically related complex exponentials), **then the coefficients are given by the formula below**.

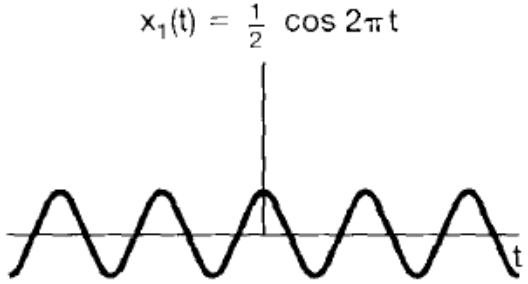
$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t},$$

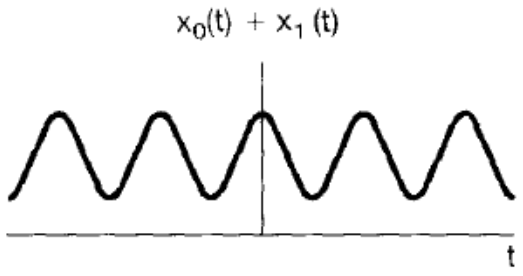
$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_T x(t)e^{-jk(2\pi/T)t}dt$$

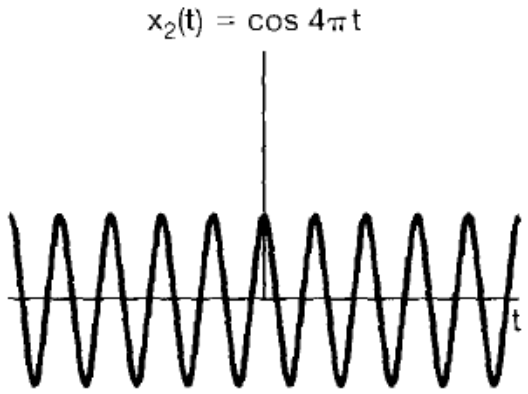
- $\{a_k\}$: *Fourier series coefficients* or the *spectral coefficients* of $x(t)$.
- a_0: the DC or constant component of $x(t)$,

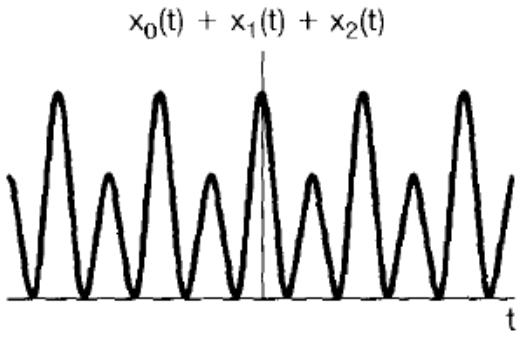
$$a_0 = \frac{1}{T}\int_T x(t)dt$$

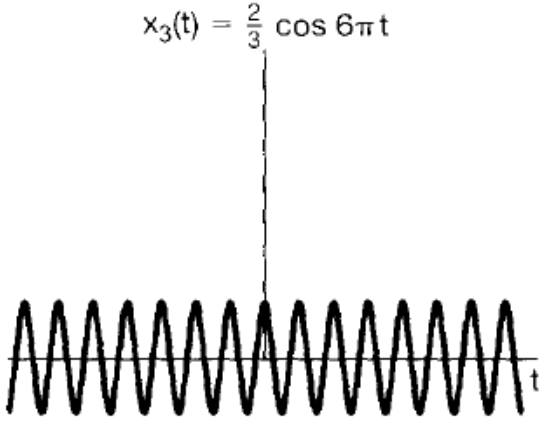
Determination of the FS Representation (CT)

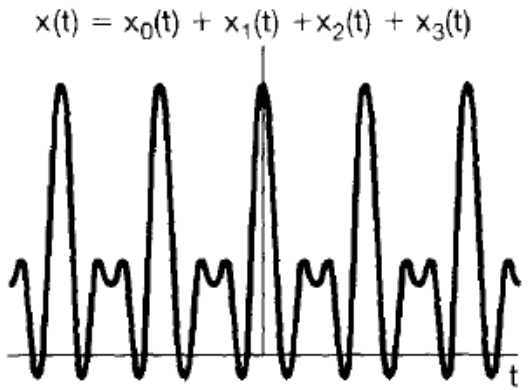












- Example: $x(t) = \sum_{k=-3}^{+3} a_k e^{jk2\pi t}$, where

  $a_0 = 1, \quad a_1 = a_{-1} = 1/4,$

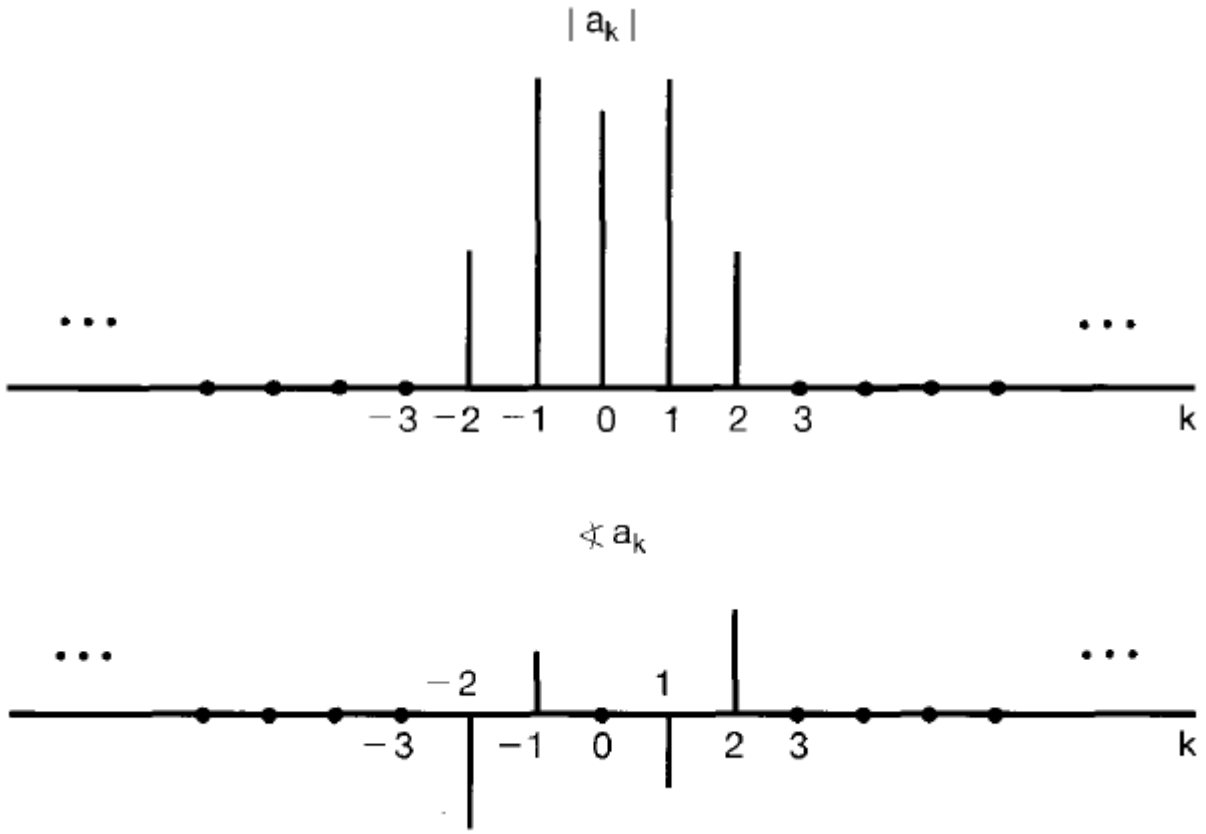
  $a_2 = a_{-2} = 1/2,$

  $a_3 = a_{-3} = 1/3$

- $x(t) = 1 + \frac{1}{4}\left(e^{j2\pi t} + e^{-j2\pi t}\right) + \frac{1}{2}\left(e^{j4\pi t} + e^{-j4\pi t}\right) + \frac{1}{3}\left(e^{j6\pi t} + e^{-j6\pi t}\right)$
- By Euler's relation, we have
- $x(t) = 1 + \frac{1}{2}cos2\pi t + \cos 4\pi t + \frac{2}{3}cos6\pi t$

Determination of the FS Representation (CT)

Q. Let $x(t) = 1 + sin\omega_0 t + 2cos\omega_0 t + \cos(2\omega_0 t + \pi/4)$ which has a fundamental frequency ω_0. Find the Fourier series coefficients.



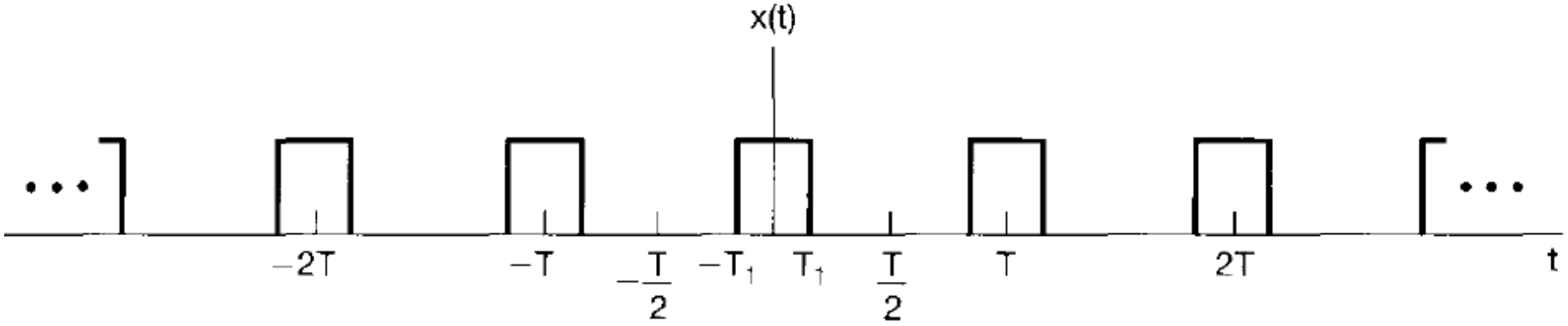
(Example 3.4)

Determination of the FS Representation (CT)

Q. The periodic square wave, shown below and defined over one period as:

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

(Example 3.5)



Determine the Fourier series coefficients for $x(t)$.

$$a_0 = \frac{1}{T}\int_{-T_1}^{T_1} dt = \frac{2T_1}{T}. \qquad a_k = \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(k\omega_0 T_1)}{k\pi}, \quad k \neq 0,$$

Fourier Series Representation of CT Periodic Signals

- **Special case: Fourier series of real periodic signals**
  - Suppose that $x(t)$ is real and can be represented by Fourier series:

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

  - With $x(t)$ being a real signal, we have

$$x(t) = x^*(t) = \sum_{k=-\infty}^{+\infty} a_k^* e^{-jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_{-k}^* e^{jk\omega_0 t}$$

  - Thus, $a_k = a_{-k}^*$; or equivalently, $a_k^* = a_{-k}$

Fourier Series Representation of CT Periodic Signals

- **Special case: Fourier series of real periodic signals**
  - By re-arranging the Fourier series representation formula, we have:

$$x(t) = a_0 + \sum_{k=1}^{+\infty} [a_k e^{jk\omega_0 t} + a_{-k} e^{-jk\omega_0 t}]$$

  - With $a_k^* = a_{-k}$, we have

$$x(t) = a_0 + \sum_{k=1}^{+\infty} [a_k e^{jk\omega_0 t} + a_k^* e^{-jk\omega_0 t}] = a_0 + \sum_{k=1}^{+\infty} 2Re\{a_k e^{jk\omega_0 t}\}$$

Fourier Series Representation of CT Periodic Signals

- **Special case: Fourier series of real periodic signals**

$$x(t) = a_0 + \sum_{k=1}^{+\infty} [a_k e^{jk\omega_0 t} + a_k^* e^{-jk\omega_0 t}] = a_0 + \sum_{k=1}^{+\infty} 2Re\{a_k e^{jk\omega_0 t}\}$$

  - Writing a_k in polar form as $r_k e^{j\theta_k}$, we have:

$$\boldsymbol{x(t) = a_0 + \sum_{k=1}^{+\infty} 2Re\{r_k e^{j(k\omega_0 t + \theta_k)}\} = a_0 + 2\sum_{k=1}^{+\infty} r_k \cos(k\omega_0 t + \theta_k)}$$

  - Writing a_k in rectangular form as $a_k = B_k + jC_k$, where B_k and C_k are real:

$$\boldsymbol{x(t) = a_0 + 2\sum_{k=1}^{+\infty} [B_k \cos(k\omega_0 t) - C_k \sin(k\omega_0 t)]}$$

Convergence of the Fourier Series

- What kind of periodic signals have Fourier series expansion?
- **Dirichlet conditions** (sufficient conditions)

  1) **Over any period, the signal must be absolutely integrable:**

$$\int_T |x(t)|dt < \infty$$

  - *Guarantees that each coefficient a_k will be finite*, since:

$$|a_k| = \left|\frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt\right| \le \frac{1}{T}\int_T \left|x(t)e^{-jk\omega_0 t}\right|dt = \frac{1}{T}\int_T |x(t)|dt < \infty$$

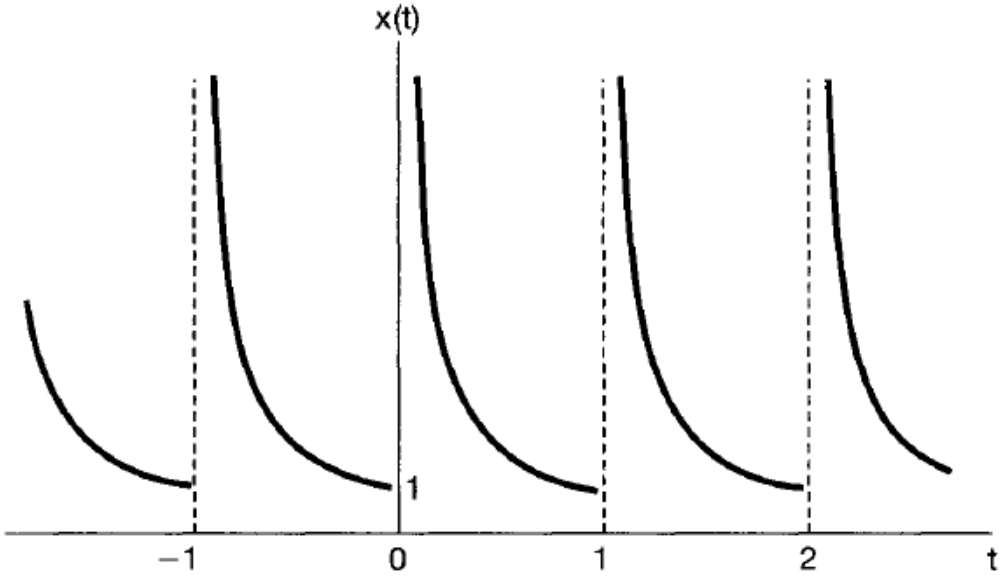
Convergence of the Fourier Series

- What kind of periodic signals have Fourier series expansion?
- **Dirichlet conditions** (sufficient conditions)

  1) **Over any period, the signal must be absolutely integrable:**

$$\int_T |x(t)|dt < \infty$$

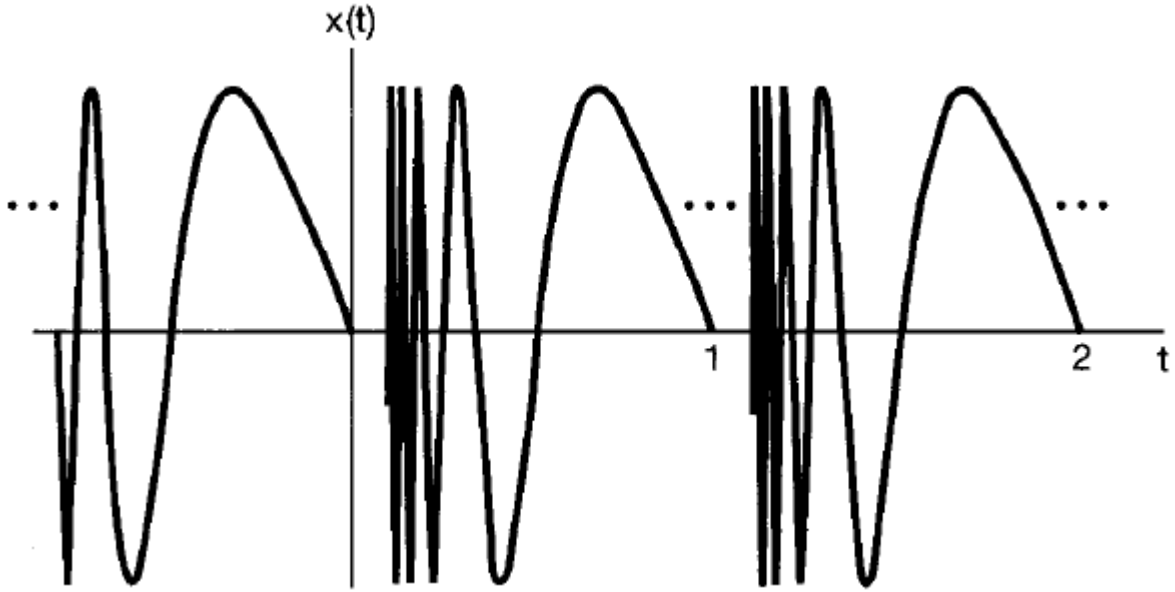
  - *Example of violation:*



$$x(t) = \frac{1}{t}, \qquad 0 < t \leq 1$$

Convergence of the Fourier Series

- What kind of periodic signals have Fourier series expansion?
- **Dirichlet conditions** (sufficient conditions)
  - **2) In any finite interval of time, $x(t)$ is of *bounded variation*;** that is, there are no more than a finite number of maxima and minima during any single period of the signal.
    - An example that meets condition 1 but not condition 2:



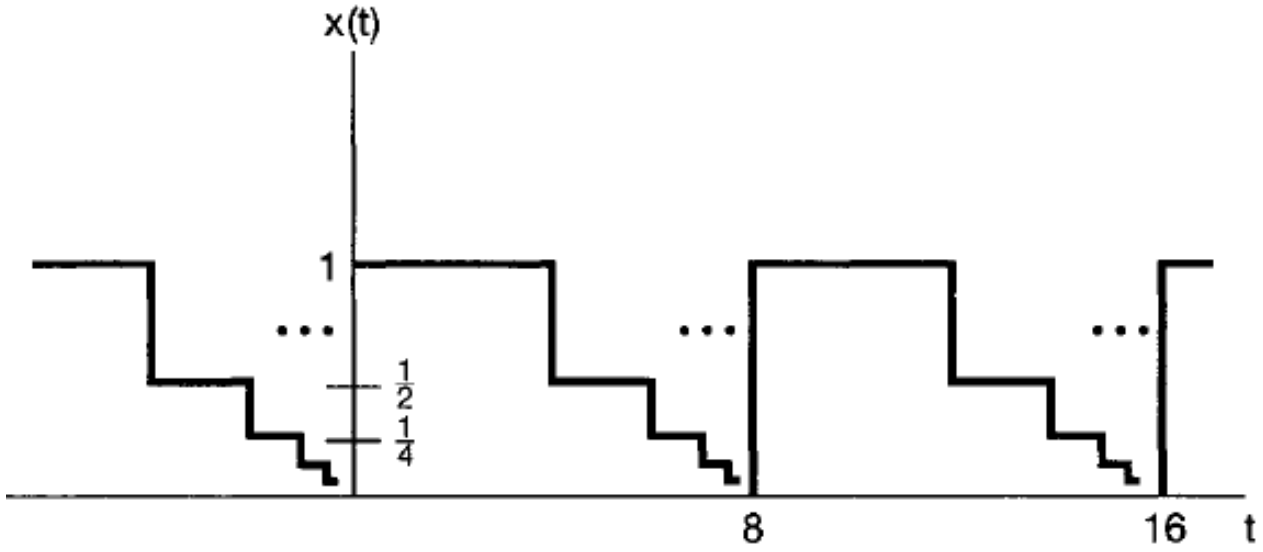
$$x(t) = \sin(\frac{2\pi}{t}), \qquad 0 < t \leq 1$$

Convergence of the Fourier Series

- What kind of periodic signals have Fourier series expansion?
- **Dirichlet conditions** (sufficient conditions)

  3) In any finite interval of time, there are only a finite number of discontinuities, and each of these discontinuities is finite.

  - An example that violates condition 3:



The signal, of period $T = 8$, is composed of an infinite number of sections, each of which is half the height and half the width of the previous section.

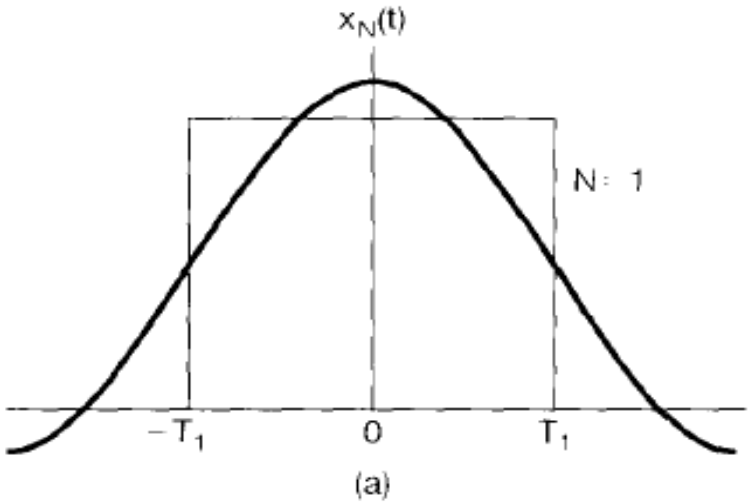
Convergence of the Fourier Series

- What kind of periodic signals have Fourier series expansion?
- **Dirichlet conditions** (sufficient conditions)

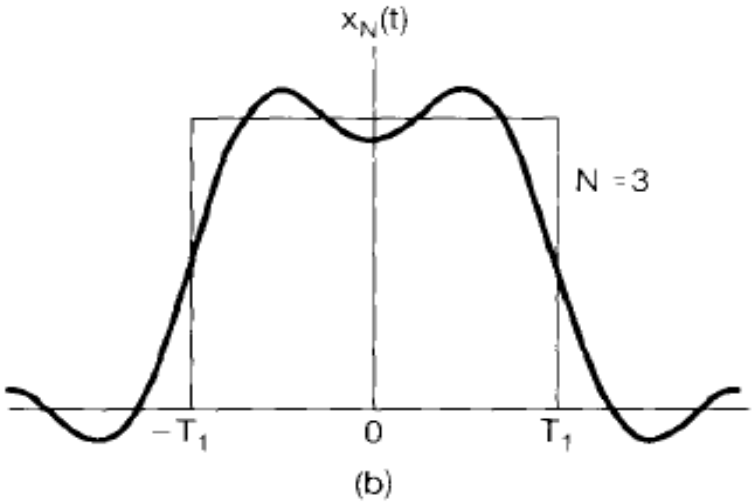
- *Signals that do not satisfy the Dirichlet conditions* are generally pathological in nature and consequently do not typically arise in practical contexts.

- *For a periodic signal that has no discontinuities*, the Fourier series representation converges and equals the original signal at every value of t.
- *For a periodic signal with finite number of discontinuities* in each period, the Fourier series representation equals the signal everywhere except *at the isolated points of discontinuity*, at which the series converges to the average value of the signal on either side of the discontinuity.

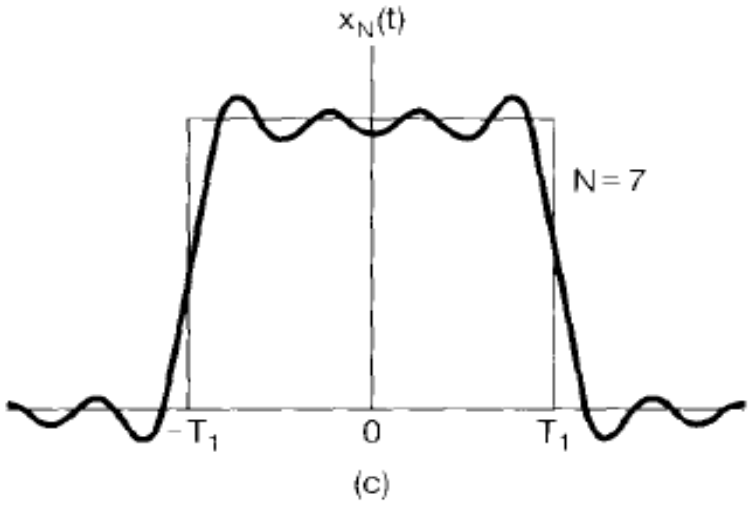
Convergence of the Fourier Series



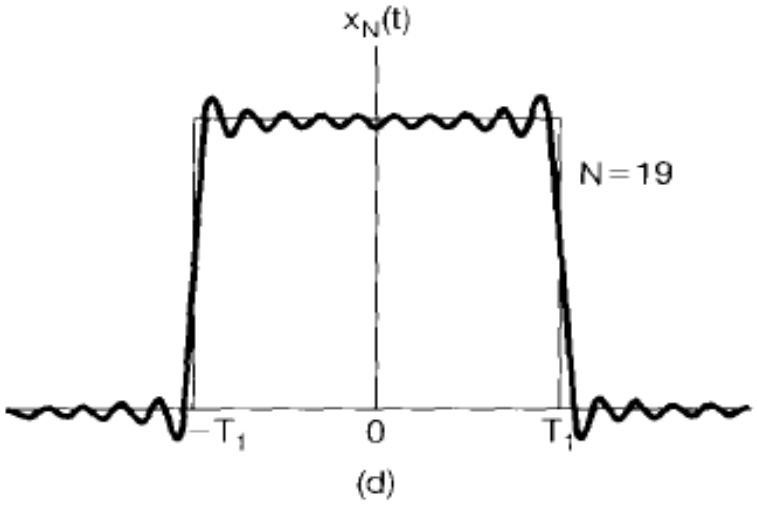
(a)



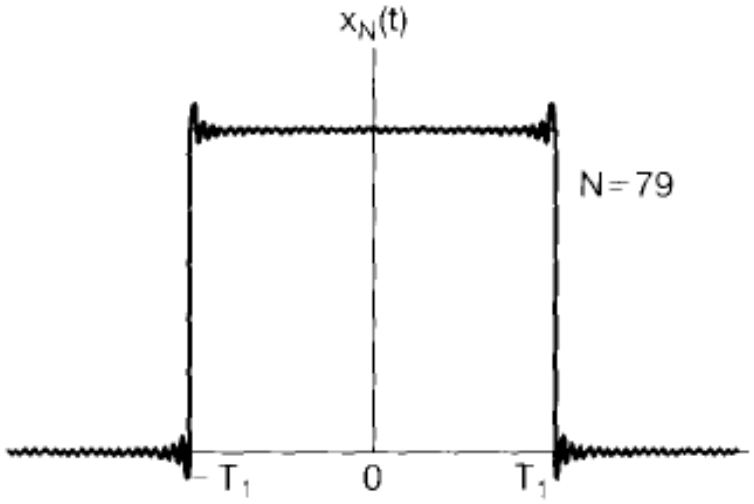
(b)



(c)



(d)



(e)

Fourier Series Representation of DT Periodic Signals

- Consider a *complex exponential signal*: $x[n] = e^{j(2\pi/N)n}$
  - Periodic with period N, fundamental frequency $\omega_0 = 2\pi/N$.
- *Recall: harmonically-related DT complex exponentials (periodic with period N)*

$$\boldsymbol{\phi_k[n] = e^{jk\omega_0 n} = e^{jk(2\pi/N)n}, \qquad k = 0, \ \pm 1, \pm 2, \ldots}$$

  - Each of these signals has a fundamental frequency: multiples of $2\pi/N$

  - *(Recall)* Are all of these signals distinct? What are the distinct ones?
    - No! $\phi_{k+N}[n] = e^{j(k+N)\omega_0 n} = e^{jk\omega_0 n} e^{jN\omega_0 n} = e^{jk\omega_0 n} = \phi_k[n]$
    - There are *only* N distinct complex exponentials in the DT harmonic family for the fundamental frequency ω_0.

Fourier Series Representation of DT Periodic Signals

- Thus, the representation of more general periodic sequences in terms of *linear combinations of the harmonically-related sequences* $\phi_k[n]$, or in other words, **the DT Fourier series for an N-periodic signal $x[n]$:**

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$$

  - where k varies over a range of N successive integers, beginning with any value k. For example, $k = 0, 1, \dots, N-1$; or $k = 3, 4, \dots, N+2$

  - This representation is called the **Discrete-Time Fourier Series**.
  - Coefficients $a_k, k \in Z$ are called the **Fourier Series coefficients**.
    - Here, a_k is also N-periodic. *i.e.,* $a_k = a_{k+N}$ for all $k \in Z$.

Determination of the FS Representation (DT)

- Consider the discrete-time Fourier series representation:

$$x[n] = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$$

- How to find the coefficients $a_k, k \in Z$?
- Multiplying both sides by $e^{-jr(2\pi/N)n}$ and summing over N terms, we have:

$$\sum_{n=<N>} x[n]e^{-jr(2\pi/N)n} = \sum_{n=<N>} \sum_{k=<N>} a_k e^{j(k-r)(2\pi/N)n}$$

- Interchanging the order of summation on the RHS, we have:

$$\sum_{n=<N>} x[n]e^{-jr(2\pi/N)n} = \sum_{k=<N>} a_k \sum_{n=<N>} e^{j(k-r)(2\pi/N)n}$$

Determination of the FS Representation (DT)

$$\sum_{n=<N>} x[n]e^{-jr(2\pi/N)n} = \sum_{k=<N>} a_k \sum_{n=<N>} e^{j(k-r)(2\pi/N)n}$$

- Suppose that $k-r$ is 0 or a multiple of N, we have:

$$\sum_{n=<N>} e^{j(k-r)(2\pi/N)n} = \sum_{n=<N>} 1 = N$$

- On the other hand, if $k-r$ is <u>*not*</u> a multiple of N, we have:

$$\sum_{n=0}^{N-1} e^{j(k-r)(2\pi/N)n} = \frac{1-e^{j(k-r)(2\pi/N)N}}{1-e^{j(k-r)(2\pi/N)}} = 0$$

Finite Sum Formula ($\alpha \neq 1$)

$$\sum_{n=0}^{N-1} \alpha^n = \frac{1-\alpha^N}{1-\alpha}$$

- Thus, $\sum_{n=<N>} x[n]e^{-jr(2\pi/N)n} = Na_r$

Determination of the FS Representation (DT)

- **Summary**: the **discrete-time Fourier series expression & the coefficients:**

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n} \quad \text{(synthesis equation)}$$

$$a_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk(2\pi/N)n} \quad \text{(analysis equation)}$$

- $\{a_k\}$: *Fourier series coefficients* or the *spectral coefficients* of $x[n]$.
  - $a_0 = \frac{1}{N} \sum_{n=<N>} x[n]$
- It is often convenient to *view a_k as a sequence defined for all values of k*, but where *only N successive* elements in the sequence will be used for FS representation.
  - $a_k = a_{k+N}$

Determination of the FS Representation (DT)

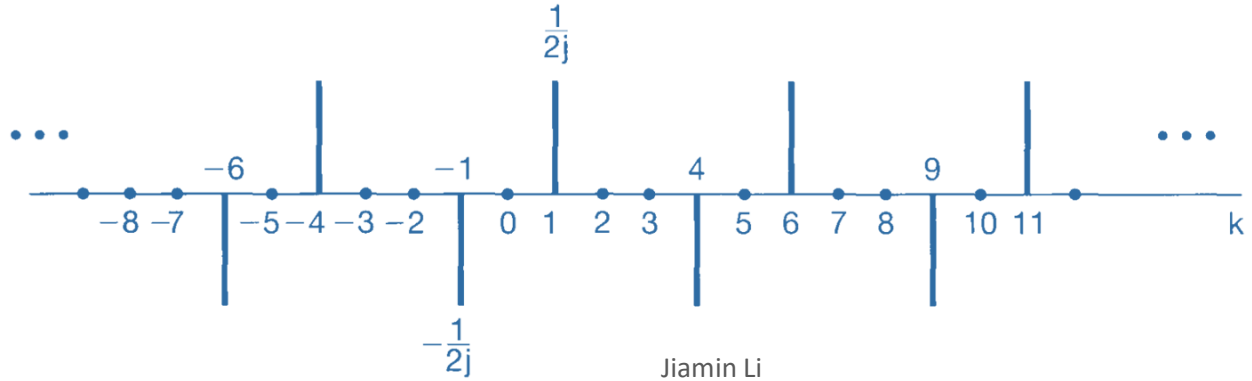
Q. Consider the signal $x[n] = sin(\omega_0 n)$. (Example 3.10)

For the case when $\omega_0 = 2\pi/N$, where N is an integer, find the FS coefficients.

$$x[n] = \frac{1}{2j}e^{j(2\pi/N)n} - \frac{1}{2j}e^{-j(2\pi/N)n}. \qquad a_1 = \frac{1}{2j}, \quad a_{-1} = -\frac{1}{2j},$$

- a_1 and a_{-1} directly observed. The remaining coefficients over the interval of summation are 0.
- These coefficients repeat with period N, but only one period is used in the synthesis equation.

For example: Fourier coefficients for $x[n] = \sin(2\pi/5)n$?



Determination of the FS Representation (DT)

(Example 3.10)

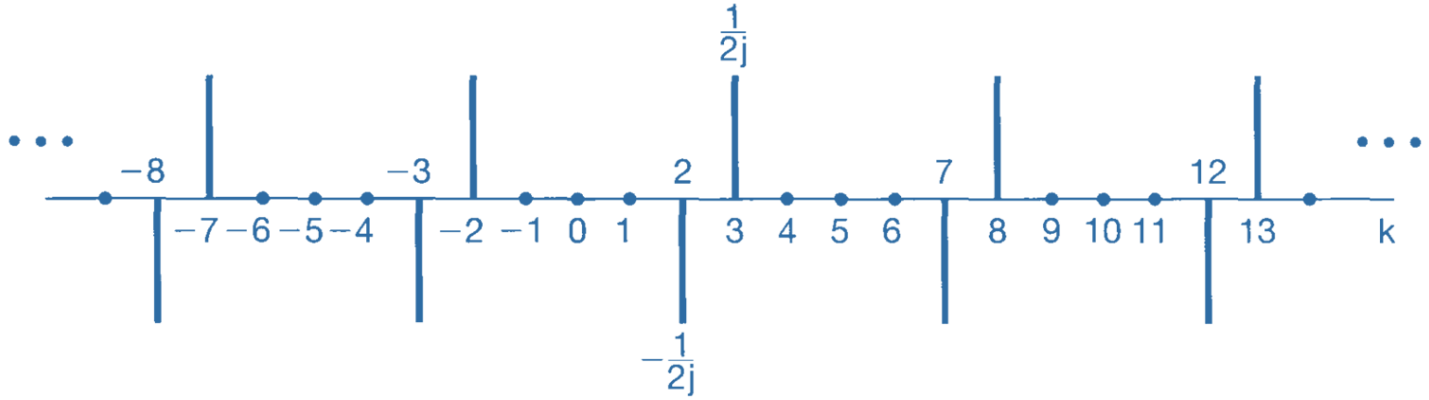
Q. Consider the signal $x[n] = sin(\omega_0 n)$.

For the case when $\omega_0 = 2\pi M/N$, where $\gcd(M, N) = 1$.

$$x[n] = \frac{1}{2j} e^{jM(2\pi/N)n} - \frac{1}{2j} e^{-jM(2\pi/N)n}, \quad a_M = (1/2j), a_{-M} = (-1/2j)$$

- a_M and a_{-M} directly observed. Other coefficients over one period of N are 0.

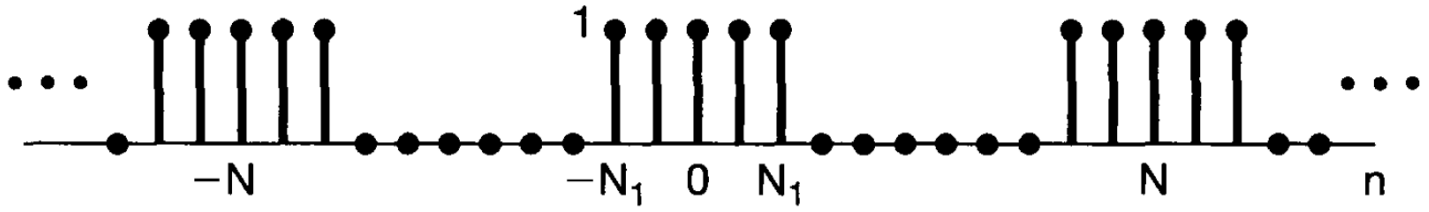
For example: Fourier coefficients for $x[n] = \sin(6\pi/5)n$?



- Note: if $x[n]$ is real, $a_k = a^*_{-k}$ for all values of k. (Analogous to CT. Proof?)

Determination of the FS Representation (DT)

Q. Determine the FS representation of the DT periodic square wave below.



(Example 3.12)

- From the analysis equation:

$$a_k = \frac{1}{N}\sum_{n=<N>} x[n]e^{-jk(2\pi/N)n} = \frac{1}{N}\sum_{n=-N_1}^{N_1} e^{-jk(2\pi/N)n}$$

Finite Sum Formula ($\alpha \neq 1$)

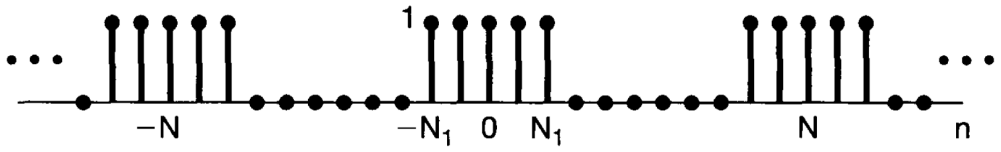
$$\sum_{n=0}^{N-1} \alpha^n = \frac{1-\alpha^N}{1-\alpha}$$

- If $k = 0$: $a_0 = (2N_1 + 1)/N$
- If $k \in \{1, 2, \dots, N-1\}$, by finite sum formula:

$$a_k = \frac{1}{N}e^{jk(2\pi/N)N_1}\left(\sum_{n=0}^{2N_1} e^{-jk(2\pi/N)n}\right) = \frac{1}{N}e^{jk(2\pi/N)N_1}\left(\frac{1-e^{-jk(2\pi/N)(2N_1+1)}}{1-e^{-jk(2\pi/N)}}\right)$$

Determination of the FS Representation (DT)

Q. Determine the FS representation of the DT periodic square wave below.



(Example 3.12)

- If $k \in \{1, 2, \dots, N-1\}$, by finite sum formula:

$$a_k = \frac{1}{N} e^{jk(2\pi/N)N_1} \left(\sum_{n=0}^{2N_1} e^{-jk(2\pi/N)n} \right) = \frac{1}{N} e^{jk(2\pi/N)N_1} \left(\frac{1 - e^{-jk(2\pi/N)(2N_1+1)}}{1 - e^{-jk(2\pi/N)}} \right)$$

- To write it in the form of trigonometric functions:

$$a_k = \frac{1}{N} \frac{e^{jk(2\pi/N)N_1} - e^{-jk(2\pi/N)(N_1+1)}}{1 - e^{-jk(2\pi/N)}} = \frac{1}{N} \frac{e^{-jk(\frac{2\pi}{N})\cdot\frac{1}{2}} \left[e^{jk\left(\frac{2\pi}{N}\right)\left(N_1+\frac{1}{2}\right)} - e^{-jk\left(\frac{2\pi}{N}\right)\left(N_1+\frac{1}{2}\right)} \right]}{e^{-jk\left(\frac{2\pi}{N}\right)\cdot\frac{1}{2}} \left[e^{jk\left(\frac{2\pi}{N}\right)\cdot\frac{1}{2}} - e^{-jk\left(\frac{2\pi}{N}\right)\cdot\frac{1}{2}} \right]}$$

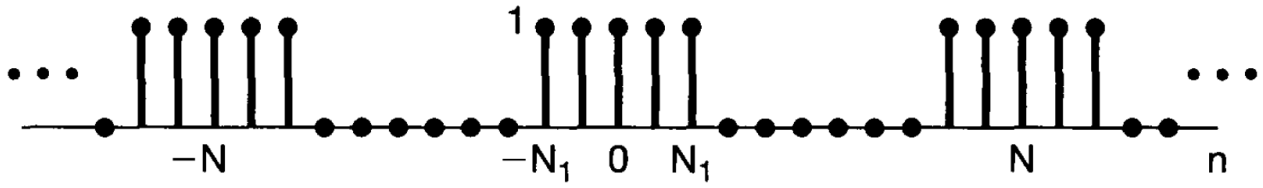
$$= \frac{1}{N} \frac{sin[2\pi k \left(N_1 + \frac{1}{2}\right)/N]}{sin(\pi k/N)}$$

DT FS Representation: Convergence?

- For FS representation of a **DT** periodic signal: No convergence issue.
  - As any DT periodic sequence $x[n]$ is completely specified by a *finite* number N of parameters, namely, the values of the sequence over one period.

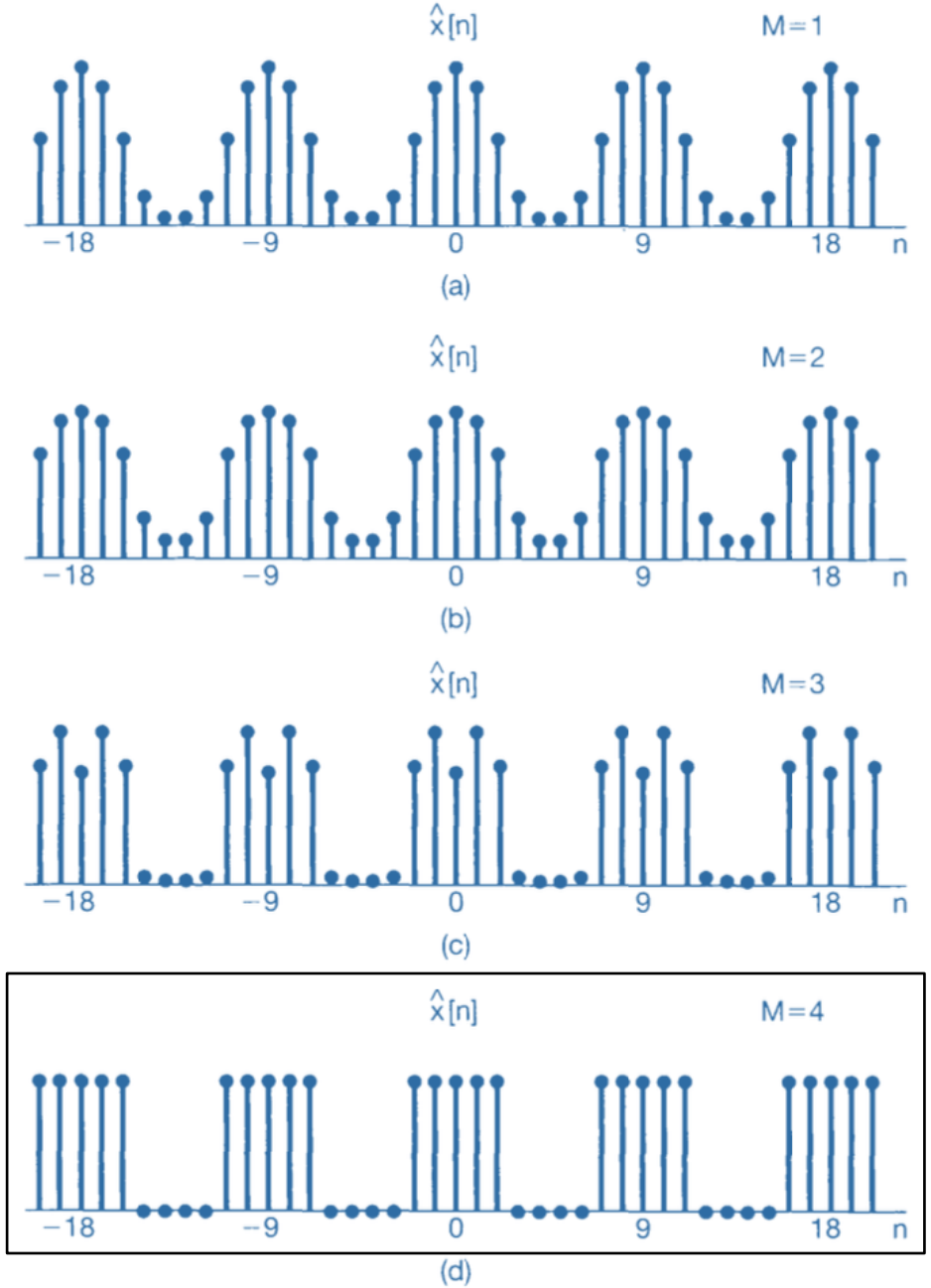
DT FS Representation: Convergence?

- Example: $N = 9, 2N_1 + 1 = 5$



- Let's depict the signals:

$$\hat{x}[n] = \sum_{k=-M}^{M} a_k e^{jk(2\pi/N)n}$$



Exactly equals $x[n]$

Properties of CT Fourier Series

- Suppose that $x(t)$ is a periodic signal with period T and fundamental frequency $\omega_0 = 2\pi/T$. Then if the Fourier series coefficients of $x(t)$ are denoted by a_k, we use the **notation** below to signify the *pairing* of a periodic signal with its Fourier series coefficients:

$$\boldsymbol{x(t) \overset{FS}{\longleftrightarrow} a_k}$$

Properties of CT Fourier Series

- **Linearity**
- Let $x(t)$ and $y(t)$ denote two periodic signals with period T and FS coefficients:

$$x(t) \overset{FS}{\longleftrightarrow} a_k$$

$$y(t) \overset{FS}{\longleftrightarrow} b_k$$

- Then,

$$\boldsymbol{z(t) = Ax(t) + By(t) \overset{FS}{\longleftrightarrow} c_k = Aa_k + Bb_k}$$

- Proof?
  - As integration is linear, this property follows directly from the definition of Fourier series coefficients.

Properties of CT Fourier Series

- **Time Shifting**
- If

$$x(t) \overset{FS}{\longleftrightarrow} a_k$$

- Then

$$\boldsymbol{x(t-t_0) \overset{FS}{\longleftrightarrow} e^{-jk\omega_0 t_0} a_k = e^{-jk(2\pi/T)t_0} a_k}$$

- Note that *when a periodic signal is shifted in time*, the *magnitude* of its *Fourier series coefficients* remain unchanged. That is $|b_k| = |a_k|$.

- Proof?

Properties of CT Fourier Series

- **Time Reversal**
- If

$$x(t) \overset{FS}{\longleftrightarrow} a_k$$

- Then

$$\boldsymbol{x(-t)} \overset{\boldsymbol{FS}}{\longleftrightarrow} \boldsymbol{a_{-k}}$$

- Proof?

- If $x(t)$ is even (*i.e.*, $x(t) = x(-t)$),
  then its Fourier series coefficients are also even (*i.e.*, $a_k = a_{-k}$).
- If $x(t)$ is odd (*i.e.*, $x(t) = -x(-t)$),
  then its Fourier series coefficients are also odd (*i.e.*, $a_k = -a_{-k}$).

Properties of CT Fourier Series

- **Time Scaling**
- If $x(t)$ is periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$,
- Then $x(\alpha t)$, where α is a positive real number, is periodic with period T/α and fundamental frequency $\alpha\omega_0$:

$$x(\alpha t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(\alpha\omega_0)t}$$

- While the *Fourier coefficients have not changed*, the Fourier series representation has changed!
  - The frequency is now $\alpha\omega_0$.
  - The harmonic family is in terms of the new period T/α.

Properties of CT Fourier Series

- **Multiplication**
- Suppose that $x(t)$ and $y(t)$ are periodic with period T and that:

$$x(t) \overset{FS}{\longleftrightarrow} a_k$$

$$y(t) \overset{FS}{\longleftrightarrow} b_k$$

- Since the product $x(t)y(t)$ is also periodic with period T:

$$\boldsymbol{x(t)y(t) \overset{FS}{\longleftrightarrow} h_k = \sum_{l=-\infty}^{\infty} a_l b_{k-l}}$$

- Proof?
- Note that the sum on the right-hand side may be interpreted as the *discrete-time convolution of the sequency a_k and b_k*.

Properties of CT Fourier Series

- **Conjugate and Conjugate Symmetry**
- If

$$x(t) \overset{FS}{\longleftrightarrow} a_k$$

- Then

Proof?

$$\boldsymbol{x^*(t) \overset{FS}{\longleftrightarrow} a^*_{-k}}$$

- If $x(t)$ is real (*i.e.*, $x(t) = x^*(t)$), then
  - $\boldsymbol{a_{-k} = a_k^*}$, $\boldsymbol{|a_{-k}| = |a_k|}$
  - $\boldsymbol{Ev\{x(t)\} \overset{FS}{\longleftrightarrow} Re\{a_k\}}$, $\boldsymbol{Od\{x(t)\} \overset{FS}{\longleftrightarrow} jIm\{a_k\}}$
- If $x(t) = x^*(t)$ and $x(t) = x(-t)$, then $\boldsymbol{a_k = a_{-k}, a_k = a_k^*}$.
- If $x(t) = x^*(t)$ and $x(t) = -x(-t)$, then $\boldsymbol{a_0 = 0, a_{-k} = -a_k, a_{-k} = a_k^*}$.

Properties of CT Fourier Series

- **Parseval's Relation** (for CT periodic signals)

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

- The *average power* in a periodic signal *equals* the *sum of the average power in all of its harmonic* components.

$$\frac{1}{T}\int_T \left|a_k e^{jk\omega_0 t}\right|^2 dt = \frac{1}{T}\int_T |a_k|^2 dt = |a_k|^2$$

- $|a_k|^2$ is the *average power* in the *kth harmonic* component of $x(t)$.

Properties of CT Fourier Series

- **Summary (Read Table 3.1 in the textbook!)**

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$x(t)$, $y(t)$ Periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$	a_k, b_k
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a^*_{-k}
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t)$, $\alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_T x(\tau)y(t-\tau)d\tau$	$Ta_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk\frac{2\pi}{T}a_k$
Integration		$\int_{-\infty}^{t} x(t)\,dt$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$a_k = a^*_{-k}$; $\mathcal{R}e\{a_k\} = \mathcal{R}e\{a_{-k}\}$; $\mathcal{I}m\{a_k\} = -\mathcal{I}m\{a_{-k}\}$; $\lvert a_k\rvert = \lvert a_{-k}\rvert$; $\measuredangle a_k = -\measuredangle a_{-k}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$x_e(t) = \mathcal{E}v\{x(t)\}$ [$x(t)$ real]; $x_o(t) = \mathcal{O}d\{x(t)\}$ [$x(t)$ real]	$\mathcal{R}e\{a_k\}$; $j\mathcal{I}m\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

Practice Example 3.6, 3.7, 3.8, 3.9 in the textbook.

Properties of DT Fourier Series

- Many properties are *analogous* to the *CT Fourier series*, with similar derivations.
- **Summary (Read Table 3.2 in the textbook!)**

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

Property	Periodic Signal	Fourier Series Coefficients
	$x[n]$, $y[n]$: Periodic with period N and fundamental frequency $\omega_0 = 2\pi/N$	a_k, b_k: Periodic with period N
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n}x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a^*_{-k}
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m}a_k$ (viewed as periodic with period mN)
Periodic Convolution	$\sum_{r=\langle N\rangle} x[r]y[n-r]$	Na_kb_k
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N\rangle} a_l b_{k-l}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running Sum	$\sum_{k=-\infty}^{n} x[k]$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{(1 - e^{-jk(2\pi/N)})}\right)a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$a_k = a^*_{-k}$; $\mathcal{R}e\{a_k\} = \mathcal{R}e\{a_{-k}\}$; $\mathcal{I}m\{a_k\} = -\mathcal{I}m\{a_{-k}\}$; $\lvert a_k\rvert = \lvert a_{-k}\rvert$; $\measuredangle a_k = -\measuredangle a_{-k}$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$x_e[n] = \mathcal{E}v\{x[n]\}$ [x[n] real]; $x_o[n] = \mathcal{O}d\{x[n]\}$ [x[n] real]	$\mathcal{R}e\{a_k\}$; $j\mathcal{I}m\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{N}\sum_{n=\langle N\rangle} |x[n]|^2 = \sum_{k=\langle N\rangle} |a_k|^2$$

Properties of DT Fourier Series

- **Multiplication**
- Suppose that $x[n]$ and $y[n]$ are periodic with period N and that:

$$x[n] \overset{FS}{\longleftrightarrow} a_k$$

$$y[n] \overset{FS}{\longleftrightarrow} b_k$$

- Since the product $x[n]y[n]$ is also periodic with period N:

$$\boldsymbol{x[n]y[n] \overset{FS}{\longleftrightarrow} d_k = \sum_{l=<N>} a_l b_{k-l}}$$

- Note that the summation variable is now *restricted* to an interval of N consecutive samples, known as ***periodic convolution*** of two periodic signals.

Properties of DT Fourier Series

- **Parseval's Relation** (for DT periodic signals)

$$\frac{1}{N}\sum_{n=<N>}|x[n]|^2 = \sum_{k=<N>}|a_k|^2$$

- The *average power* in a periodic signal *equals* the *sum of the average power in all of its harmonic* components.
- $|a_k|^2$ is the *average power* in the *kth harmonic* component of $x[n]$, similarly to the CT case.

Properties of DT Fourier Series

Q. Suppose we are given the following facts about a sequence $x[n]$:

1. $x[n]$ is periodic with period $N = 6$.
2. $\sum_{n=0}^{5} x[n] = 2$.
3. $\sum_{n=2}^{7} (-1)^n x[n] = 1$.
4. $x[n]$ has the minimum power per period among the set of signals satisfying the above conditions.

Determine the sequence $x[n]$.

Example 3.14

Fourier Series & LTI Systems

- Recall:

$$x(t) = e^{st} \rightarrow \boxed{\text{LTI System } h(t)} \rightarrow y(t) = H(s)e^{st}$$

$$H(s) = \int_{-\infty}^{+\infty} h(\tau)e^{-s\tau}d\tau$$

$$x(t) = \sum_k a_k e^{s_k t}$$

$$y(t) = \sum_k a_k H(s_k) e^{s_k t}$$

$$x[n] = z^n \rightarrow \boxed{\text{LTI System } h[n]} \rightarrow y(t) = H(z)z^n$$

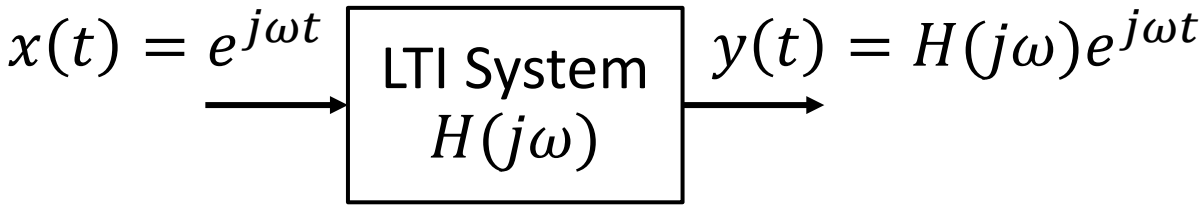
$$H(z) = \sum_{k=-\infty}^{+\infty} h[k]z^{-k}$$

$$x[n] = \sum_k a_k {z_k}^n$$

$$y[n] = \sum_k a_k H(z_k) {z_k}^n$$

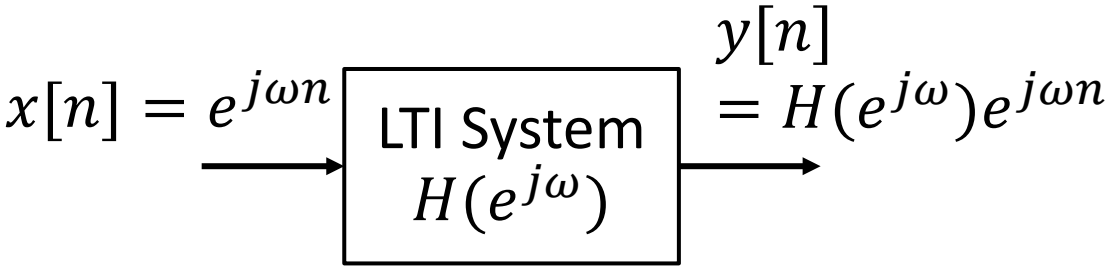
Fourier Series & LTI Systems

- Recall: frequency response of an LTI system



$(s = j\omega)$

$$H(j\omega) = \int_{-\infty}^{+\infty} h(\tau)e^{-j\omega\tau}d\tau$$



$(z = e^{j\omega})$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} h[k]e^{-j\omega k}$$

Fourier Series & LTI Systems

- **Fourier series and LTI systems: CT**

$$x(t) = \sum_k a_k e^{jk\omega_0 t} \rightarrow \boxed{\begin{array}{c}\text{LTI System} \\ h(t)\end{array}} \rightarrow y(t) = \sum_k H(jk\omega_0) a_k e^{jk\omega_0 t}$$

$$(s = j\omega)$$

$$a_k \longrightarrow \underbrace{H(jk\omega_0)}_{\text{“gain”}} a_k$$

$$H(jk\omega_0) = |H(jk\omega_0)| e^{j\angle H(jk\omega_0)}$$

Including both amplitude & phase

- The effect of the LTI system is to modify each a_k through multiplication by the value of the frequency response at the corresponding frequency.

Fourier Series & LTI Systems

- **Fourier series and LTI systems: DT**

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} \rightarrow \boxed{\begin{array}{c}\text{LTI System} \\ h[n]\end{array}} \rightarrow y[n] = \sum_{k=<N>} H(e^{jk\omega_0}) a_k e^{jk\omega_0 n}$$

$$(s = e^{j\omega})$$

$$a_k \longrightarrow \underbrace{H(e^{jk\omega_0})}_{\text{“gain”}} a_k$$

$$H(e^{jk\omega_0}) = |H(e^{jk\omega_0})| e^{j\angle H(e^{jk\omega_0})}$$

Including both amplitude & phase

- The effect of the LTI system is to modify each a_k through multiplication by the value of the frequency response at the corresponding frequency.

Summary

- **Response of LTI systems to complex exponentials: CT & DT**
  - System response calculation with transfer function
  - Special case: periodic complex exponentials & system frequency response
- **Fourier series representation of CT periodic signals**
  - Fourier series representation & Fourier coefficients calculation
  - Special case: *real* periodic signals, two trigonometrical representations
  - Convergence issues
  - Properties
- **Fourier series representation of DT periodic signals**
  - Fourier series representation & Fourier coefficients calculation
  - Properties (analogy and differences compared to CT)
- **LTI system frequency response of Fourier series: CT & DT**

Practice Example 3.16, 3.17 in the textbook